Status	Finished
Started	Saturday, 31 October 2026, 8:52 PM
Completed	Saturday, 31 October 2026, 8:56 PM
Duration	3 mins 44 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. The parts are ignored
- b. The parts are merged without sorting
- c. Each part is sorted independently
- d. Each part is searched for a specific element

The correct answer is: Each part is sorted independently

Question **2**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Deleting data from a list
- b. Arranging data in a particular format
- c. Searching for data in a list
- d. Inserting data into a list

The correct answer is: Arranging data in a particular format

Question **3**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It complicates the analysis process
- b. It has no effect on the analysis process
- c. It allows for quicker identification of duplicates
- d. It makes the analysis slower

The correct answer is: It allows for quicker identification of duplicates

Question **4**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Makes data harder to manage
b. Decreases the efficiency of algorithms
c. Provides a basis for other algorithms to work efficiently
d. Makes code execution slower

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Bubble Sort
b. Merge Sort
c. Linear Search
d. Binary Search

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Searching
b. Complexity
c. Sorting
d. Merging

The correct answer is:
Complexity

Question **7**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Binary Search
- b. Linear Search
- c. Merge Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **8**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It is based on the divide-and-conquer approach
- b. It is less efficient than bubble sort
- c. It sorts data using a single pass
- d. It does not require recursion

The correct answer is: It is based on the divide-and-conquer approach

Question **9**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Heap Sort
- b. Bubble Sort
- c. Merge Sort
- d. Quick Sort

The correct answer is: Bubble Sort

Question **10**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It makes the search operation slower
- b. It increases the number of comparisons needed
- c. It allows for faster searching
- d. It has no effect on the search operation

The correct answer is: It allows for faster searching

Question **11**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It divides the list into sublists
- b. It recursively sorts subproblems
- c. It bubbles up the largest element in each pass
- d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass

Question **12**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Duplicates
- b. Searching
- c. Selection
- d. All of the above

The correct answer is: All of the above

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Insertion Sort
- b. Merge Sort
- c. Quick Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Merge Sort
- b. Linear Search
- c. Binary Search
- d. Bubble Sort

The correct answer is: Bubble Sort

Question **15**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It is less efficient than custom sorting algorithms
- b. It requires external libraries
- c. It sorts data out of the box efficiently
- d. It only works with integer arrays

The correct answer is: It sorts data out of the box efficiently